# Numerical Analysis

## Lecture2: Solutions of Equations in One Variable

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Special Thanks to Prof. Guanding Yu

# Outline

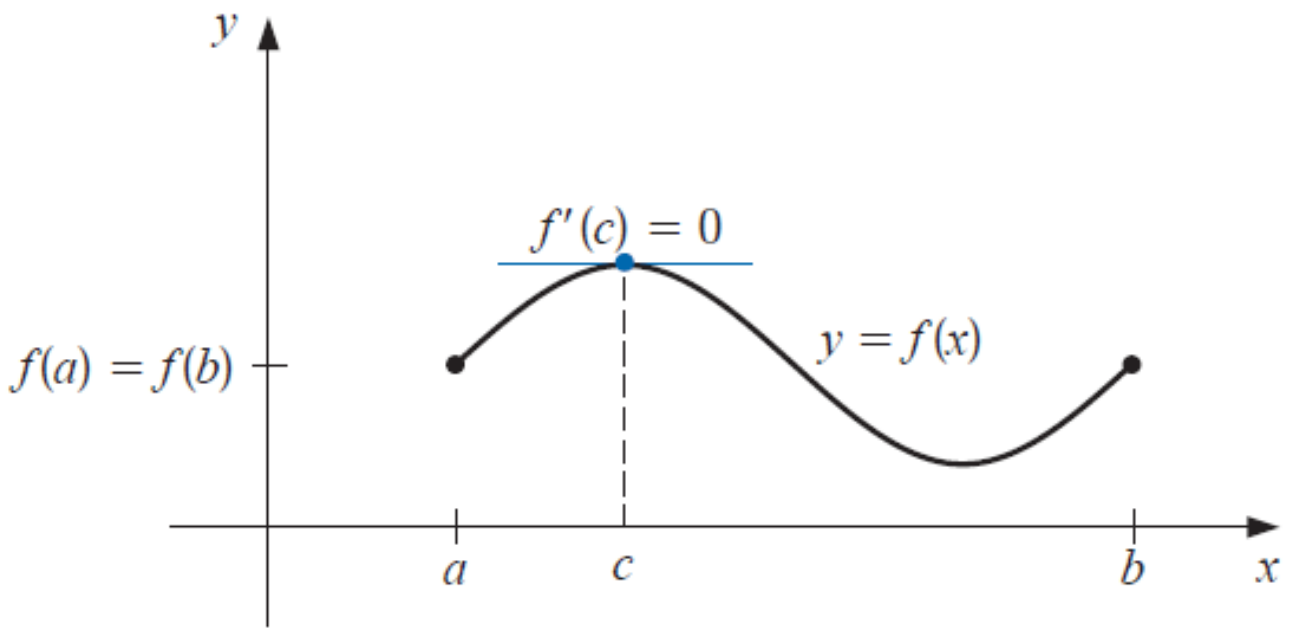
1 The Root-Finding Problem

2 Newton's Method

3 Error Analysis for Iterative Methods

## Rolle's Theorem

Suppose $f \in C[a,b]$ and $f$ is differentiable on $(a,b)$. If $f(a) = f(b)$, then a number $c$ in $(a,b)$ exists with $f'(c) = 0$.

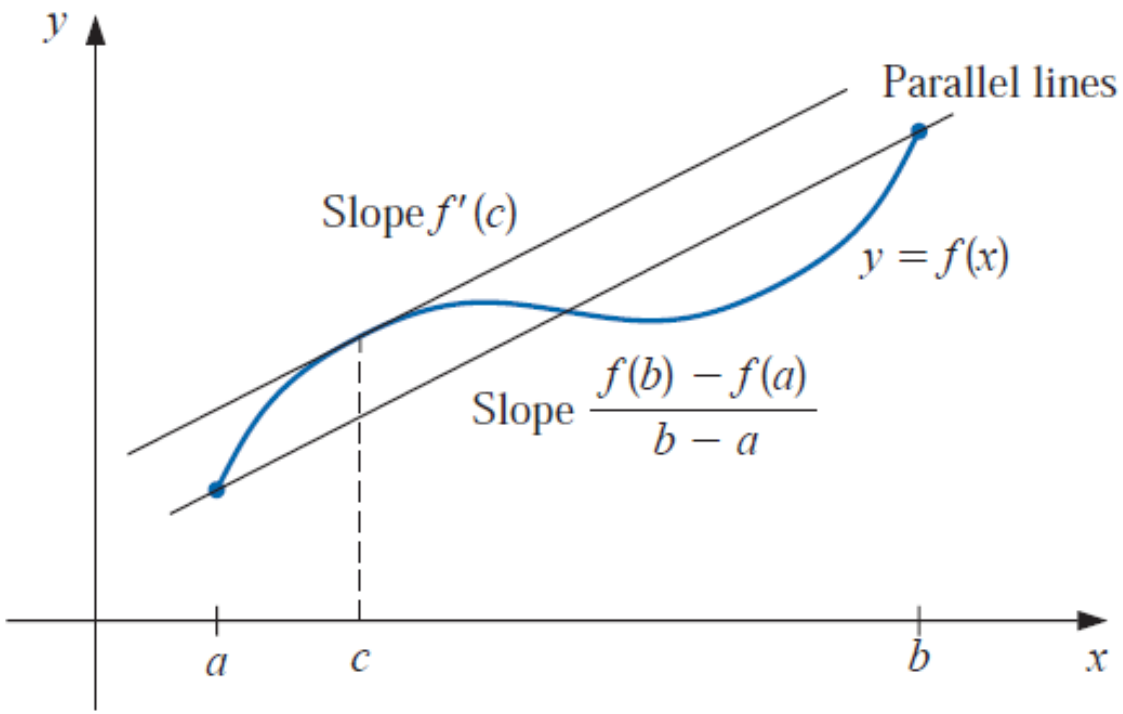


# Review of Calculus

## Mean Value Theorem

If $f \in C[a, b]$ and $f$ is differentiable on $(a, b)$, then a number $c$ in $(a, b)$ exists with

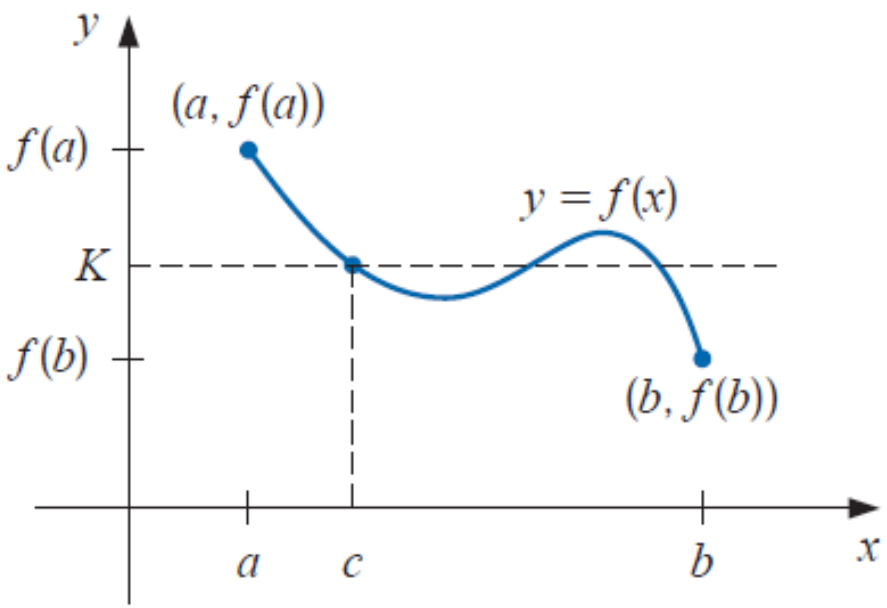
$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$



# Review of Calculus

## Intermediate Value Theorem

If $f \in C[a, b]$ and $K$ is any number between $f(a)$ and $f(b)$, then there exists a number $c$ in $(a, b)$ for which $f(c) = K$.



# The Root-Finding Problem

## The Root-Finding problem

- This process involves finding a root, or solution, of an equation of the form

$$f(x) = 0$$

  for a given function $f$.
- A root of this equation is also called a zero of the function $f$.

# The Bisection Method

## Assumptions

- Suppose $f \in C[a,b]$ and $f(a) \cdot f(b) < 0$.
- By the IVT, there exists an $x$ in $(a,b)$ with $f(x) = 0$.
- We assume for simplicity that the root in this interval is unique.

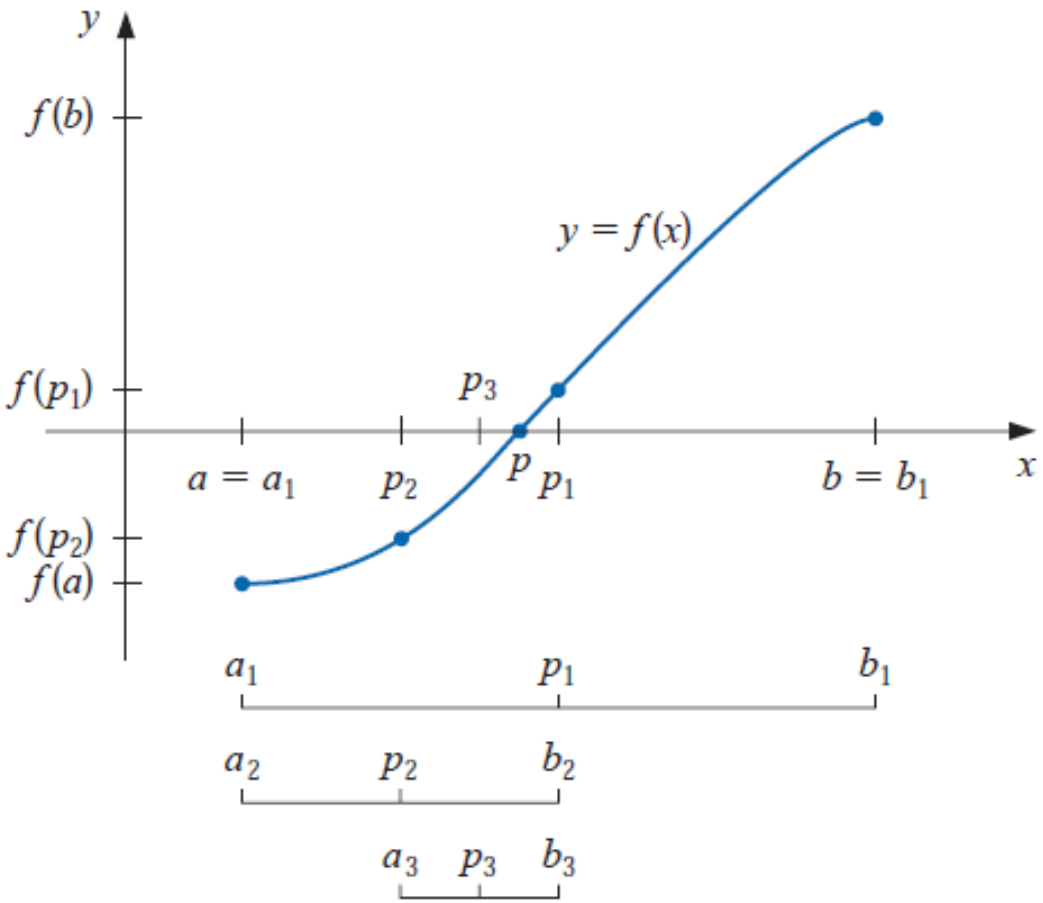
## Solution - The Bisection Method

- Divide the interval $[a,b]$ by computing the midpoint

$$p = (a+b)/2$$

- If $f(p)$ has same sign as $f(a)$, consider new interval $[p,b]$.
- If $f(p)$ has same sign as $f(b)$, consider new interval $[a,p]$.
- Repeat until interval small enough to approximate $x$ well.

# The Bisection Method



# The Bisection Method

## Algorithm

INPUT endpoints $a, b$; tolerance $TOL$; maximum number of iterations $N_0$.

OUTPUT approximate solution $p$ or message of failure.

*Step 1* Set $i = 1$;
$FA = f(a)$.

*Step 2* While $i \leq N_0$ do Steps 3–6.

*Step 3* Set $p = a + (b - a)/2$; (*Compute* $p_i$.)
$FP = f(p)$.

*Step 4* If $FP = 0$ or $(b - a)/2 < TOL$ then
OUTPUT $(p)$; (*Procedure completed successfully.*)
STOP.

*Step 5* Set $i = i + 1$.

*Step 6* If $FA \cdot FP > 0$ then set $a = p$; (*Compute* $a_i, b_i$.)
$FA = FP$
else set $b = p$. (*FA is unchanged.*)

*Step 7* OUTPUT ('Method failed after $N_0$ iterations, $N_0 =$', $N_0$);
(*The procedure was unsuccessful.*)
STOP.

# Convergence

## Theorem

*Suppose that* $f \in C[a,b]$ *and* $f(a) \cdot f(b) < 0$. *The Bisection method generates a sequence* $\{p_n\}_{n=1}^{\infty}$ *approximating a zero* $p$ *of* $f$ *with*

$$|p_n - p| \leq \frac{b-a}{2^n}, \quad \text{when } n \geq 1.$$

## Proof

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## Proof

- For each $n \geq 1$, we have

$$b_n - a_n = \frac{1}{2^{n-1}}(b-a) \quad \text{and} \quad p \in (a_n, b_n).$$

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- For each $n \geq 1$, we have
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- Since $p_n = \frac{1}{2}(a_n + b_n)$ for all $n \geq 1$, it follows that
$$|p_n - p| \leq \frac{1}{2}(b_n - a_n) = \frac{b-a}{2^n}.$$

# Rate of Convergence

## Theorem

*Suppose that* $f \in C[a,b]$ *and* $f(a) \cdot f(b) < 0$. *The Bisection method generates a sequence* $\{p_n\}_{n=1}^{\infty}$ *approximating a zero* $p$ *of* $f$ *with*

$$|p_n - p| \leq \frac{b-a}{2^n}, \quad \text{when } n \geq 1.$$

## Theorem

*The sequence* $\{p_n\}_{n=1}^{\infty}$ *converges to* $p$ *with rate of convergence* $O(1/2^n)$:

$$p_n = p + O\left(\frac{1}{2^n}\right).$$

# Remarks

## Remarks

- The Bisection Method has a number of significant drawbacks.
  1. It is very slow to converge in that $N$ may be quite large before $|p - p_N|$ becomes sufficiently small.
  2. It is possible that a good intermediate approximation may be inadvertently discarded.
- It will always converge to a solution however and, for this reason, is often used to provide a good initial approximation for a more efficient procedure.

# Example

## Example 1

Show that $f(x) = x^3 + 4x^2 - 10 = 0$ has a root in $[1, 2]$, and use the Bisection method to determine an approximation to the root that is accurate to at least within $10^{-4}$.

| $n$ | $a_n$ | $b_n$ | $p_n$ | $f(p_n)$ |
|---|---|---|---|---|
| 1 | 1.0 | 2.0 | 1.5 | 2.375 |
| 2 | 1.0 | 1.5 | 1.25 | −1.79687 |
| 3 | 1.25 | 1.5 | 1.375 | 0.16211 |
| 4 | 1.25 | 1.375 | 1.3125 | −0.84839 |
| 5 | 1.3125 | 1.375 | 1.34375 | −0.35098 |
| 6 | 1.34375 | 1.375 | 1.359375 | −0.09641 |
| 7 | 1.359375 | 1.375 | 1.3671875 | 0.03236 |
| 8 | 1.359375 | 1.3671875 | 1.36328125 | −0.03215 |
| 9 | 1.36328125 | 1.3671875 | 1.365234375 | 0.000072 |
| 10 | 1.36328125 | 1.365234375 | 1.364257813 | −0.01605 |
| 11 | 1.364257813 | 1.365234375 | 1.364746094 | −0.00799 |
| 12 | 1.364746094 | 1.365234375 | 1.364990235 | −0.00396 |
| 13 | 1.364990235 | 1.365234375 | 1.365112305 | −0.00194 |

# The Fixed-Point Problem

## The Root-Finding Problem

Given a function $f(x)$ where $a \leq x \leq b$, find values $p$ such that

$$f(p) = 0$$

## The Fixed-Point Problem

Given such a function, $f(x)$, we now construct an auxiliary function $g(x)$ such that

$$p = g(p)$$

whenever $f(p) = 0$.

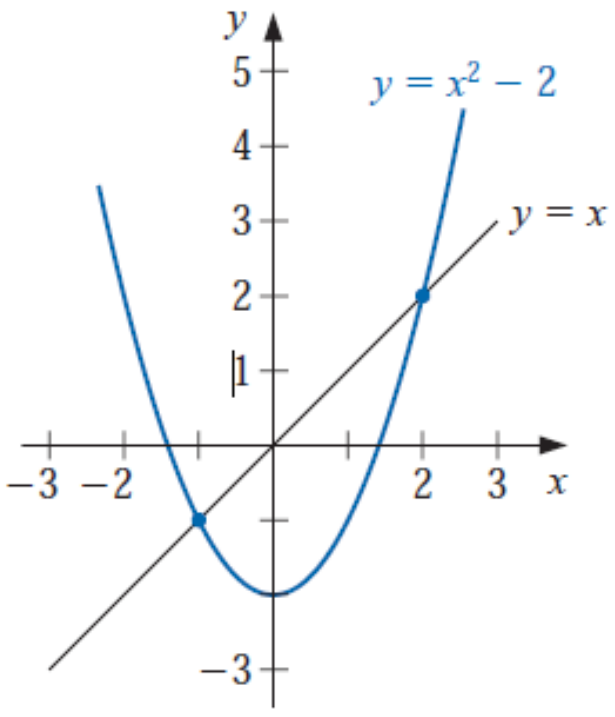
- This construction is not unique.
- The problem of finding $p$ such that $p = g(p)$ is known as the fixed point problem.

# The Fixed-Point Problem

## A Fixed Point

If $g$ is defined on $[a, b]$ and $g(p) = p$ for some $p \in [a, b]$, then the function $g$ is said to have the fixed point p in $[a, b]$.

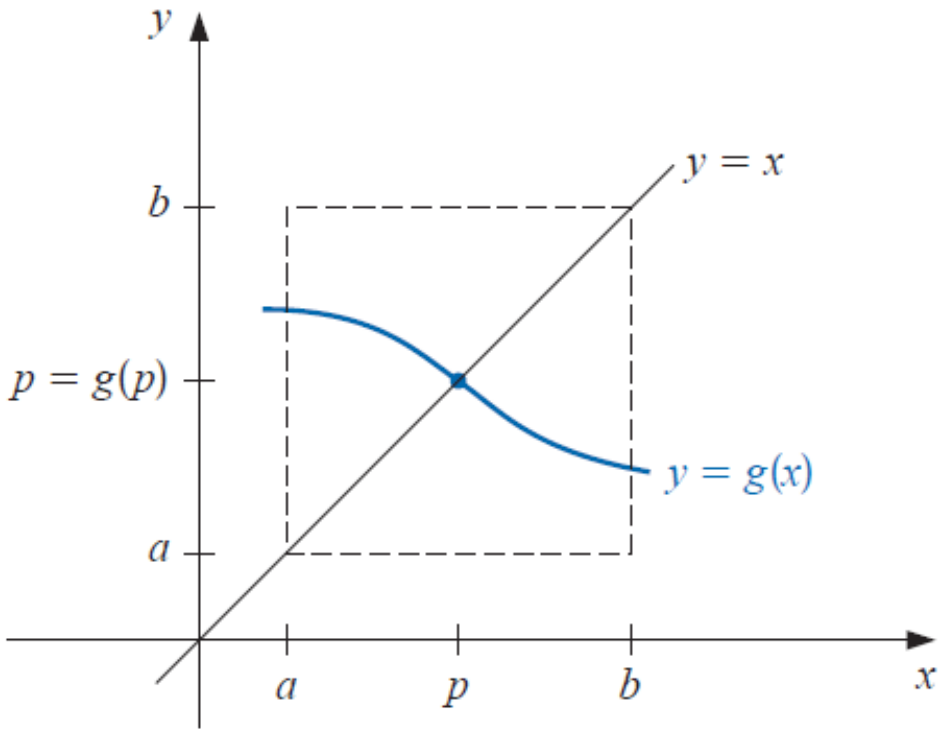
Ex: Determine any fixed points of the function $g(x) = x^2 - 2$.



# Existence of Fixed Points

## Theorem (Existence of Fixed Points)

*If $g \in C[a,b]$ and $g(x) \in [a,b]$ for all $x \in [a,b]$ then the function $g$ has a fixed point in $[a,b]$.*



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## Proof

- If $g(a) = a$ or $g(b) = b$, the existence of a fixed point is obvious.
- If not, then it must be true that $g(a) > a$ and $g(b) < b$.

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- If not, then it must be true that $g(a) > a$ and $g(b) < b$.
- Define $h(x) = g(x) - x$; $h$ is continuous on $[a,b]$ and, moreover,

$$h(a) = g(a) - a > 0, \qquad h(b) = g(b) - b < 0.$$

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- The IVT implies that there exists $p \in (a,b)$ for which $h(p) = 0$.

# Uniqueness of Fixed Points

## Theorem (Uniqueness of Fixed Points)

*Let* $g \in C[a,b]$ *and* $g(x) \in [a,b]$ *for all* $x \in [a,b]$. *Further if* $g'(x)$ *exists on* $[a,b]$ *and*

$$|g'(x)| \leq k < 1, \forall x \in [a,b],$$

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- Suppose that $p$ and $q$ are both fixed point in $[a,b]$ with $p \neq q$.

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## Proof

- Suppose that $p$ and $q$ are both fixed point in $[a,b]$ with $p \neq q$.
- By the MVT, a number $\xi$ exists between $p$ and $q$ in $[a,b]$ with

$$\begin{aligned} |p-q| = |g(p)-g(q)| &= |g'(\xi)||p-q| \\ &\leq k|p-q| < |p-q| \end{aligned}$$

which is a contradiction

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  which is a contradiction
- Hence, $p = q$ and the fixed point in $[a,b]$ is unique.

# Fixed-Point Iteration

## A Method to Solve the Fixed-Point Problem

- Choose an initial approximation $p_0$ and generate the sequence $\{p_n\}_{n=0}^{\infty}$ by letting $p_n = g(p_{n-1})$, for each $n \geq 1$.

# Fixed-Point Iteration

## A Method to Solve the Fixed-Point Problem

- Choose an initial approximation $p_0$ and generate the sequence $\{p_n\}_{n=0}^{\infty}$ by letting $p_n = g(p_{n-1})$, for each $n \geq 1$.
- If the sequence converges to $p$ and $g$ is continuous, then

$$p = \lim_{n\to\infty} p_n = \lim_{n\to\infty} g(p_{n-1}) = g\left(\lim_{n\to\infty} p_{n-1}\right) = g(p),$$

and a solution to $x = g(x)$ is obtained.

# Fixed-Point Iteration

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- If the sequence converges to $p$ and $g$ is continuous, then

$$p = \lim_{n\to\infty} p_n = \lim_{n\to\infty} g(p_{n-1}) = g\left(\lim_{n\to\infty} p_{n-1}\right) = g(p),$$

  and a solution to $x = g(x)$ is obtained.
- This technique is called fixed-point iteration.

# Fixed-Point Iteration

## Fixed-Point Algorithm

INPUT initial approximation $p_0$; tolerance $TOL$; maximum number of iterations $N_0$.

OUTPUT approximate solution $p$ or message of failure.

*Step 1* Set $i = 1$.

*Step 2* While $i \leq N_0$ do Steps 3–6.

*Step 3* Set $p = g(p_0)$. (*Compute $p_i$.*)

*Step 4* If $|p - p_0| < TOL$ then
OUTPUT $(p)$; (*The procedure was successful.*)
STOP.

*Step 5* Set $i = i + 1$.

*Step 6* Set $p_0 = p$. (*Update $p_0$.*)

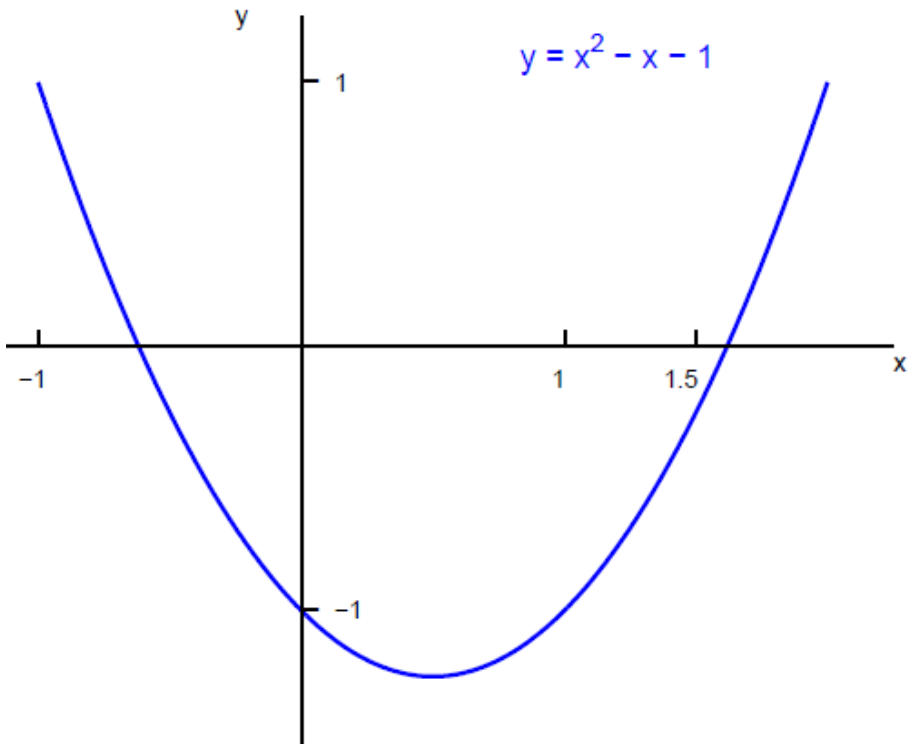
*Step 7* OUTPUT ('The method failed after $N_0$ iterations, $N_0 =$', $N_0$);
(*The procedure was unsuccessful.*)
STOP.

# Examples

## Example 1

Find the positive root for the quadratic equation:

$$x^2 - x - 1 = 0$$



# Example 1

## Solution 1

Convert the quadratic equation $f(x) = x^2 - x - 1 = 0$ to a fixed-point problem.

- Transpose the equation $f(x) = 0$ for variable $x$:

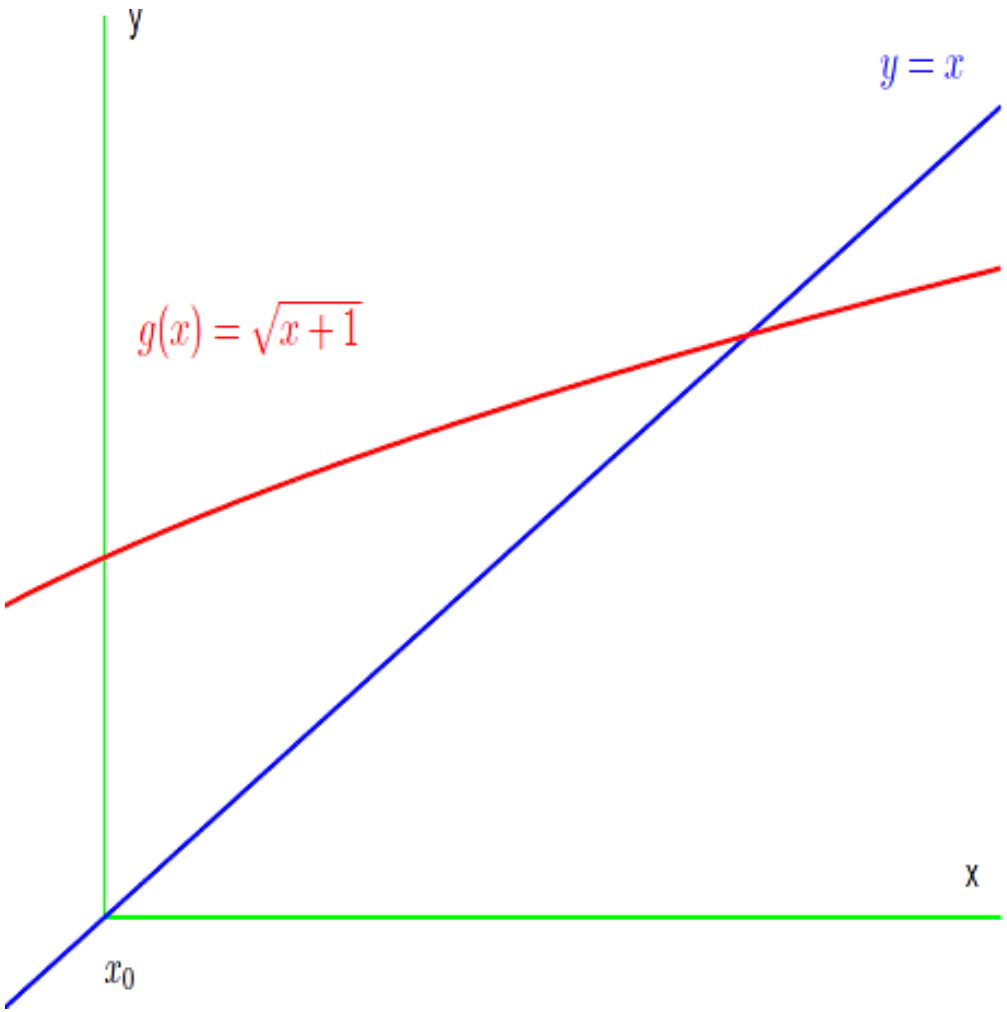
$$\begin{aligned} x^2 - x - 1 &= 0 \\ \Rightarrow \quad x^2 &= x + 1 \\ \Rightarrow \quad x &= \pm\sqrt{x+1} \end{aligned}$$

- The constructed fixed-point problem:

$$g(x) = \sqrt{x+1}$$

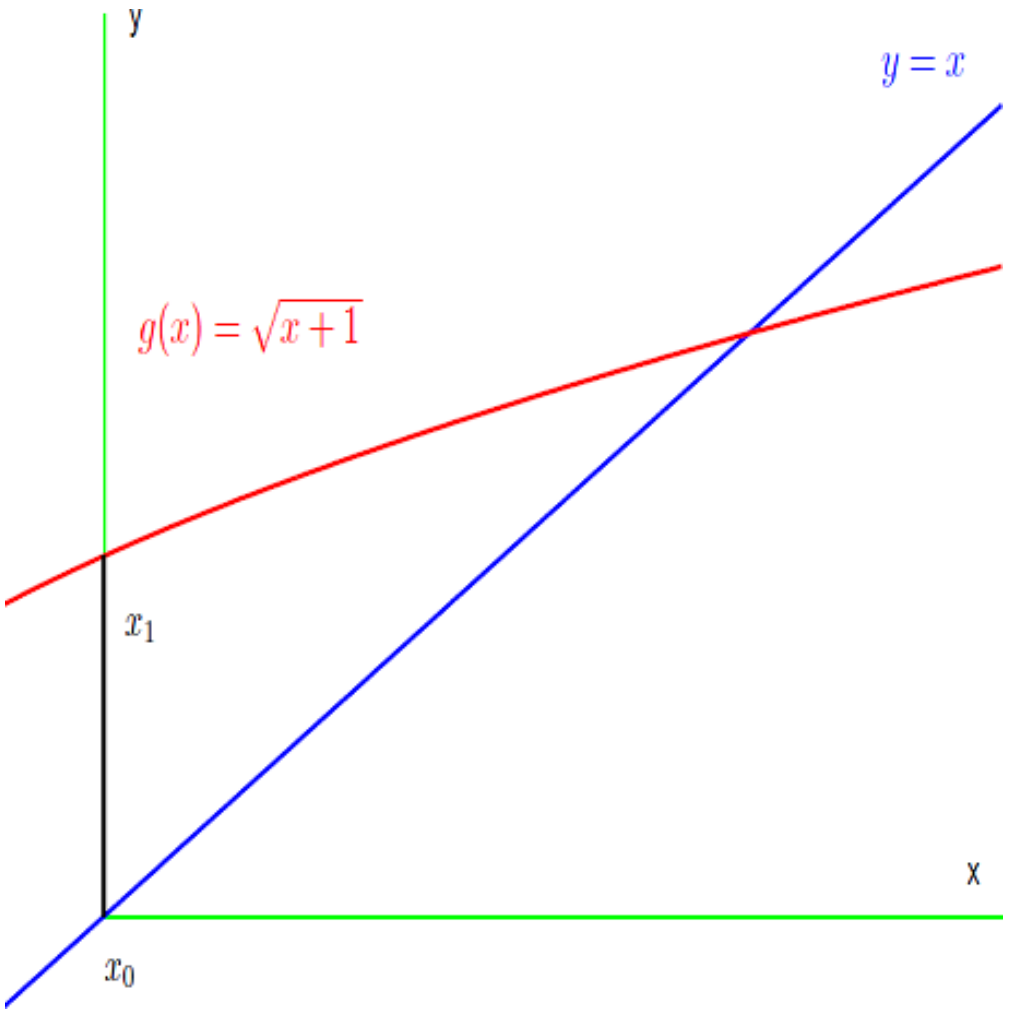
# Example 1

## Solution 1: $x_{n+1} = g(x_n) = \sqrt{x_n + 1}$ with $x_0 = 0$



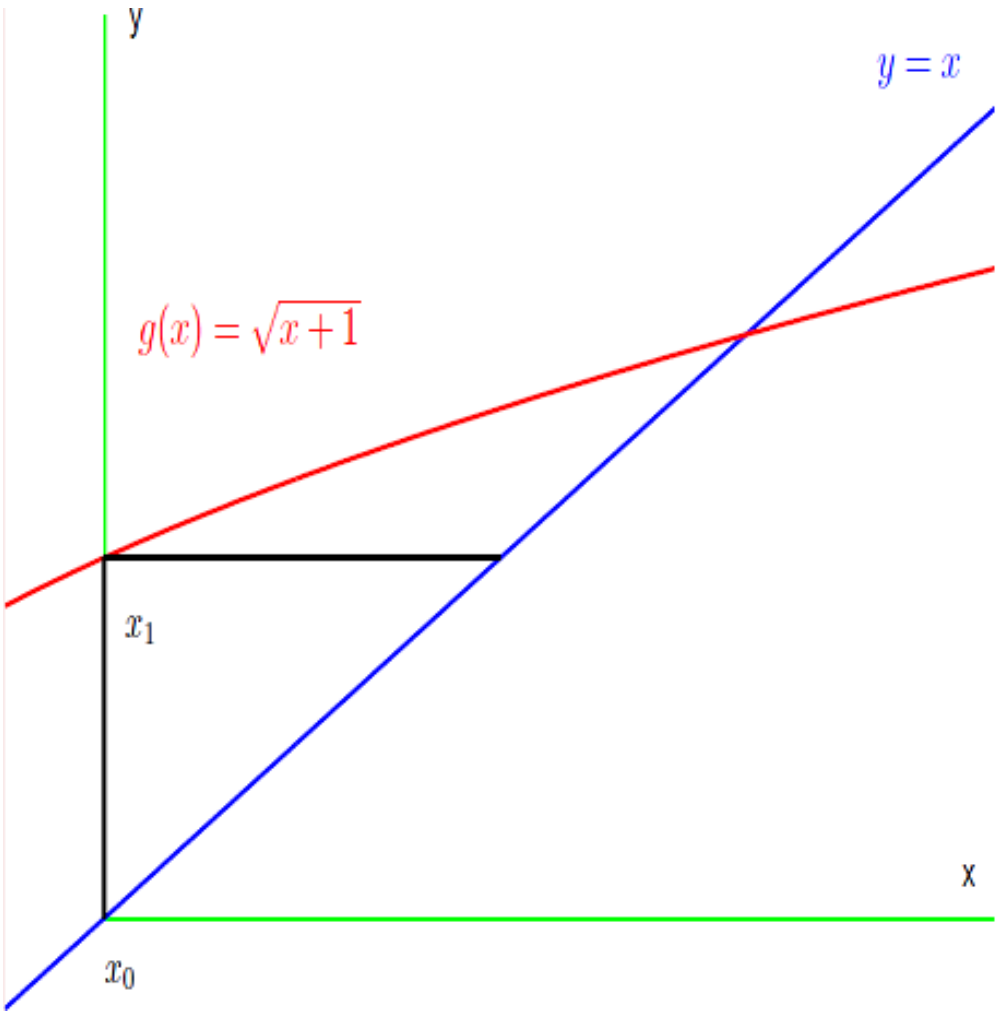
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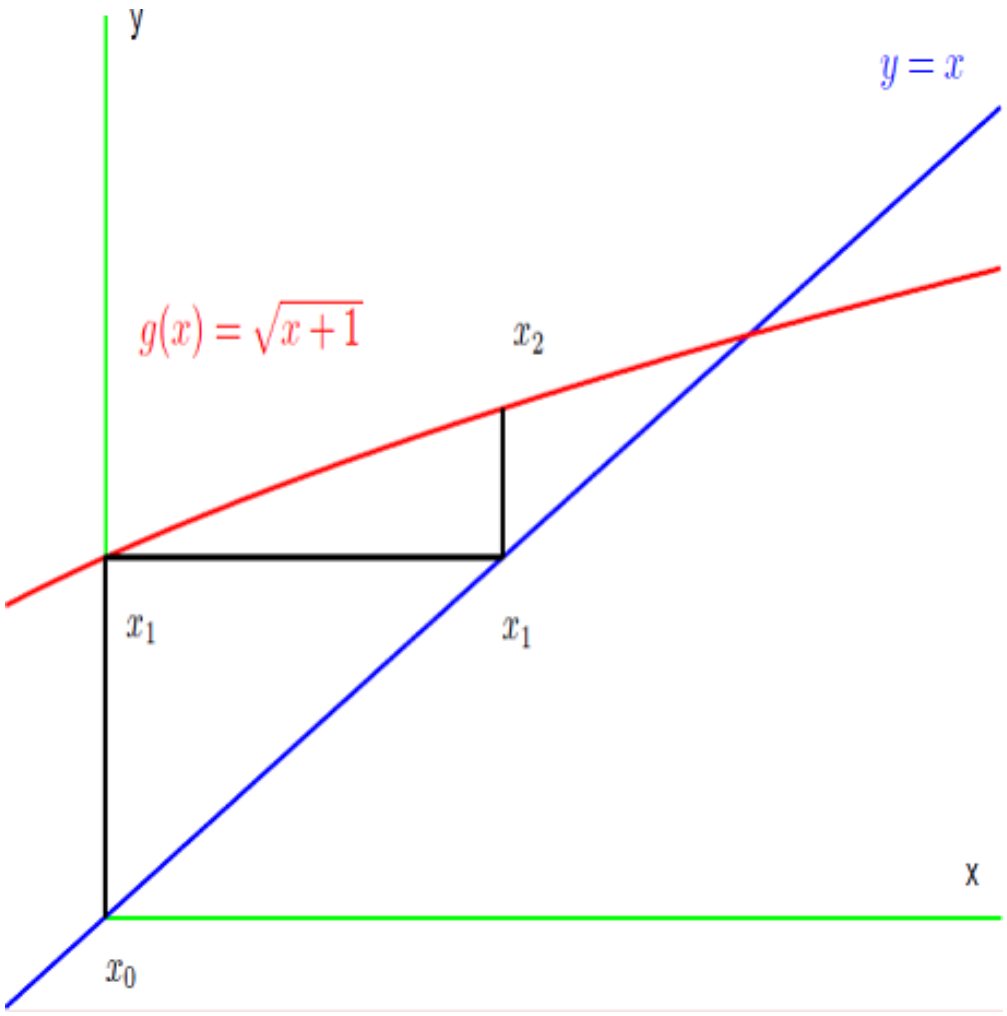
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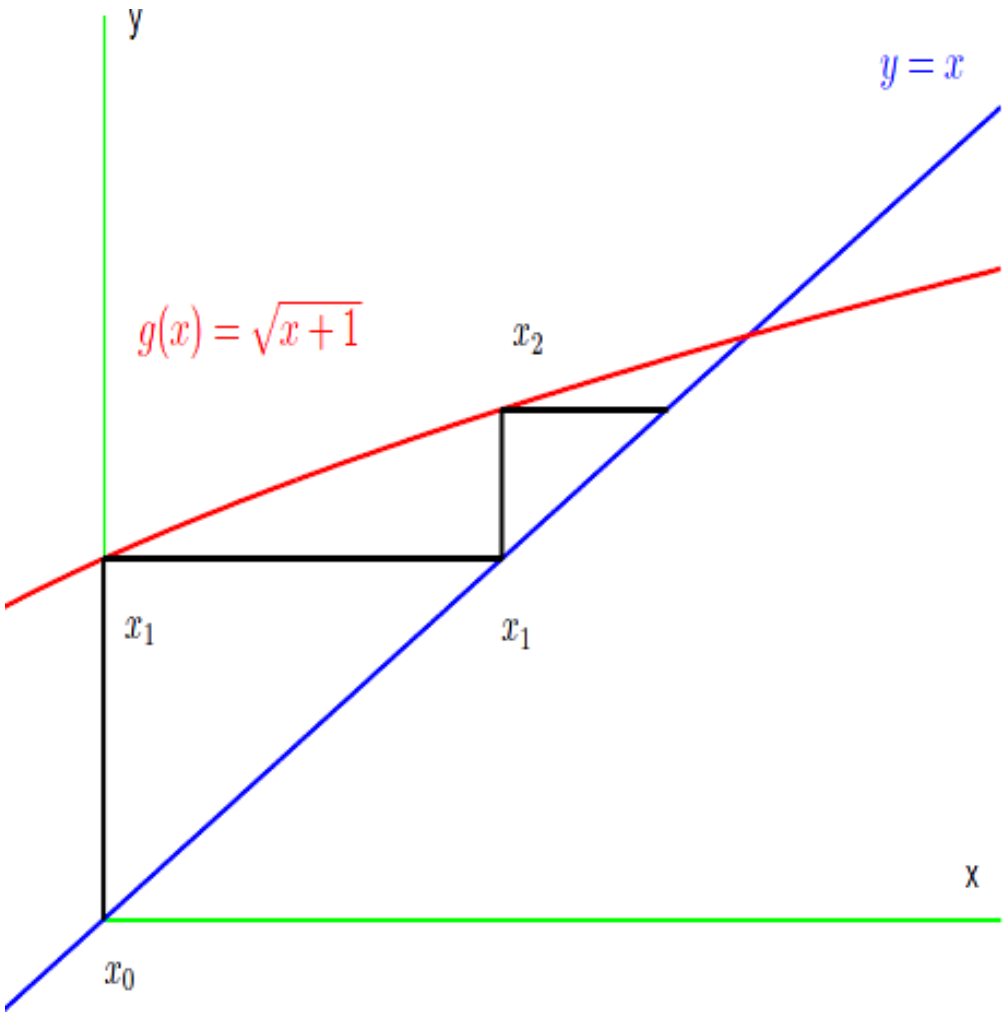
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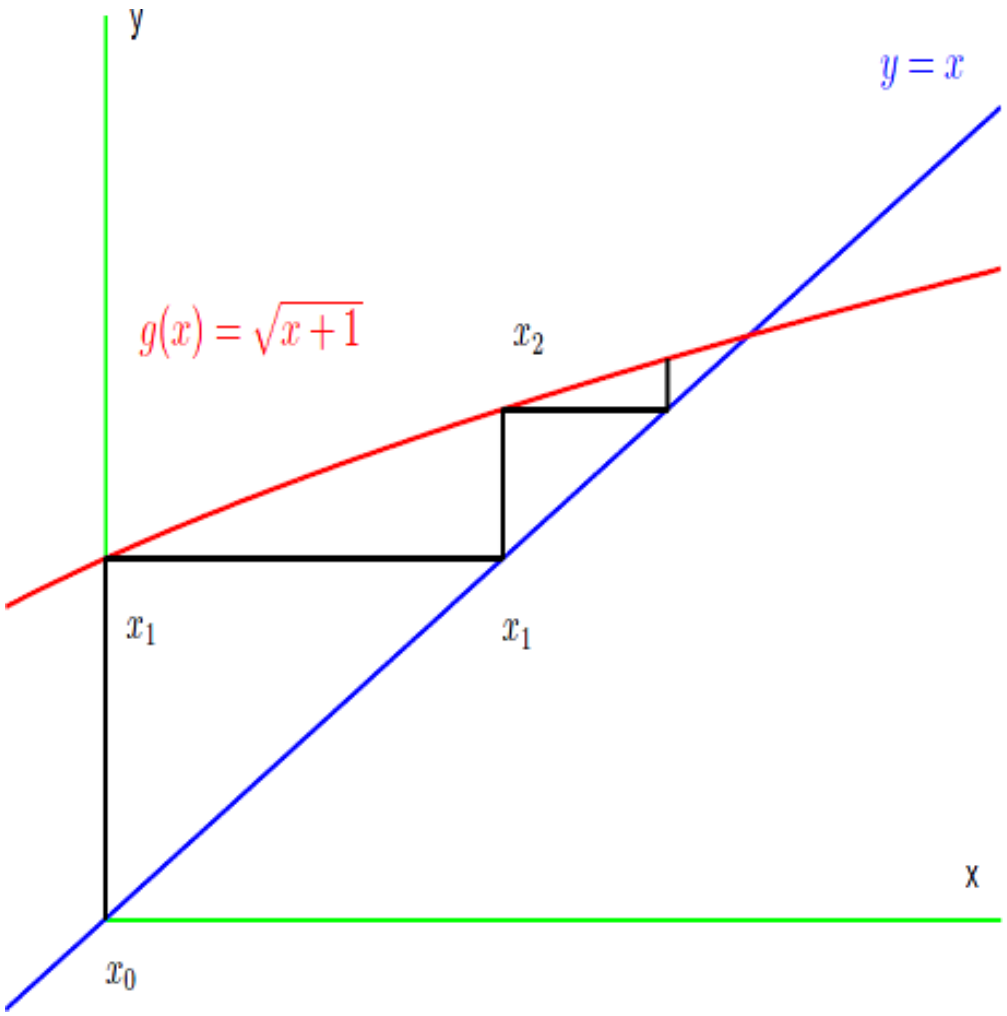
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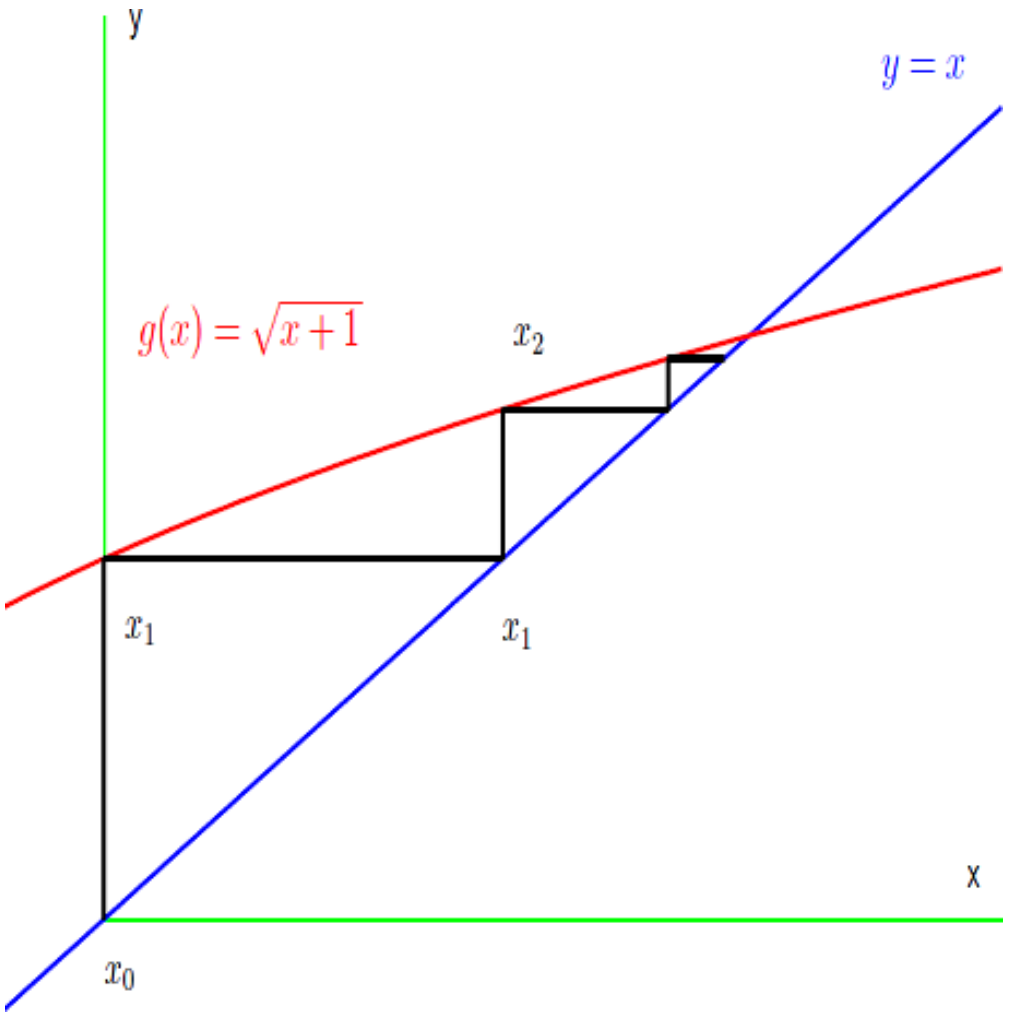
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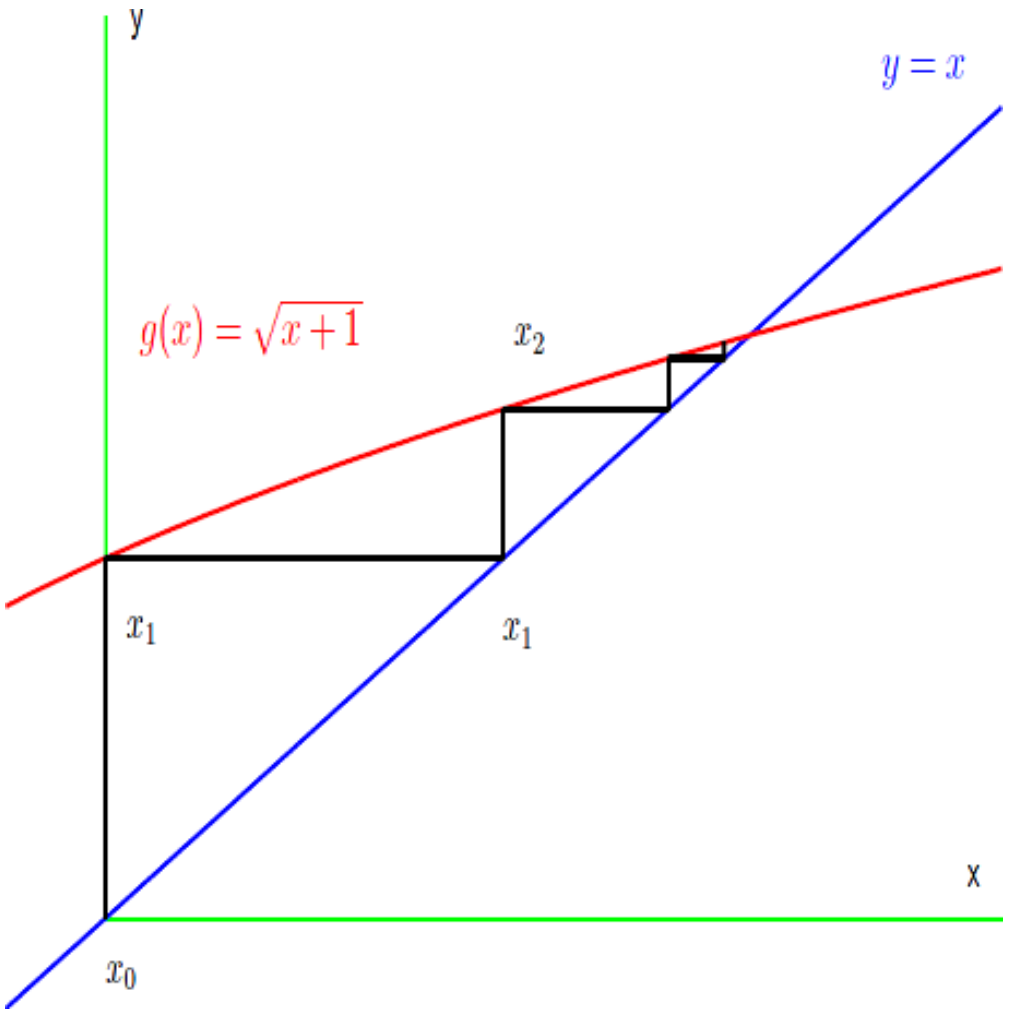
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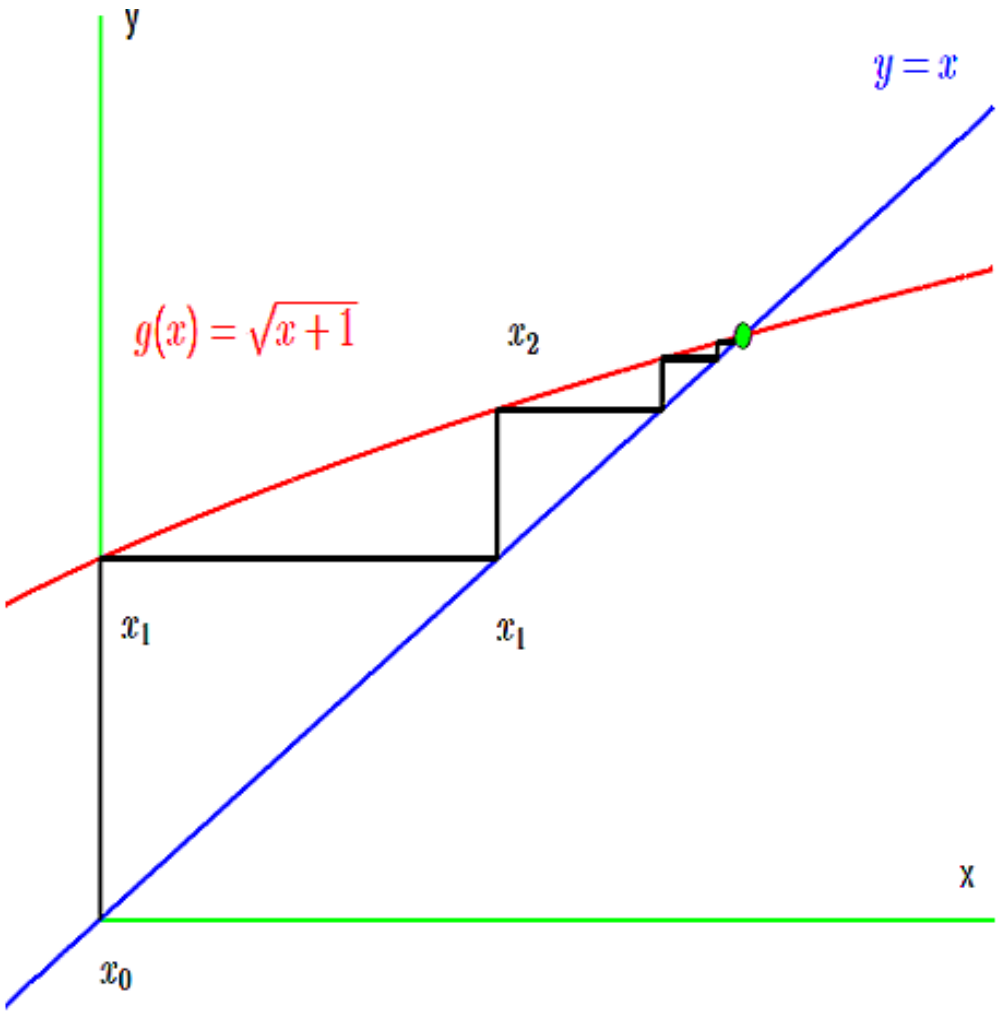
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# Example 1

## Solution 2

Convert the quadratic equation $f(x) = x^2 - x - 1 = 0$ to a fixed-point problem.

- Transpose the equation $f(x) = 0$ for variable $x$:

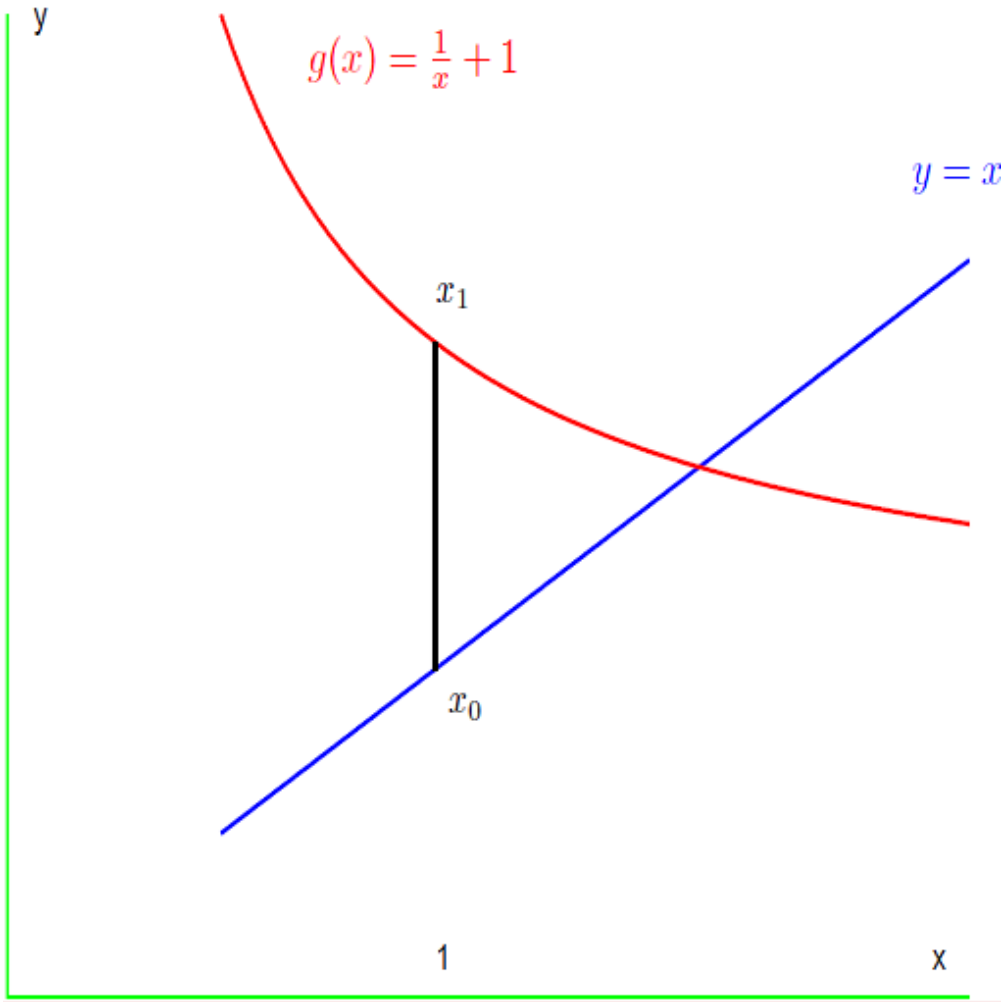
$$\begin{aligned} x^2 - x - 1 &= 0 \\ \Rightarrow \quad x^2 &= x + 1 \\ \Rightarrow \quad x &= 1 + \frac{1}{x} \end{aligned}$$

- The constructed fixed-point problem:

$$g(x) = 1 + \frac{1}{x}$$

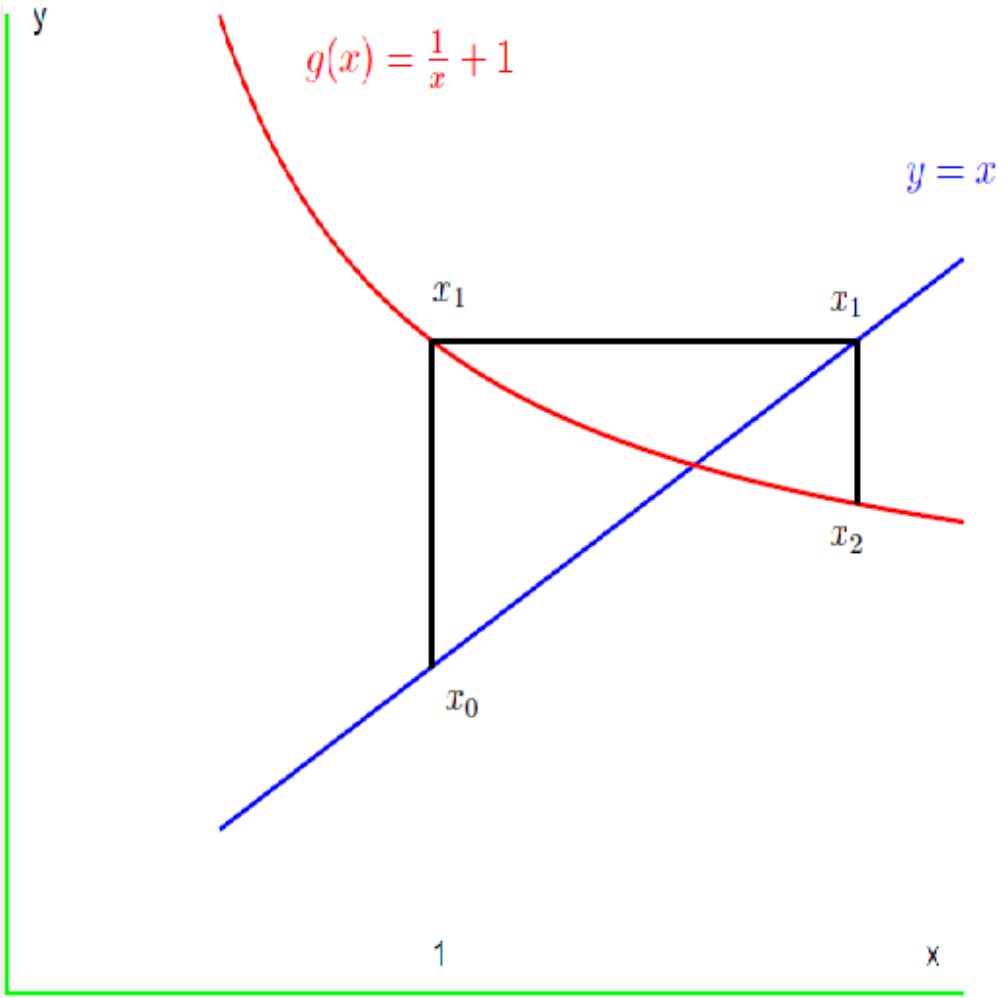
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## Solution 2: $x_{n+1} = g(x_n) = 1 + \frac{1}{x_n}$ with $x_0 = 1$



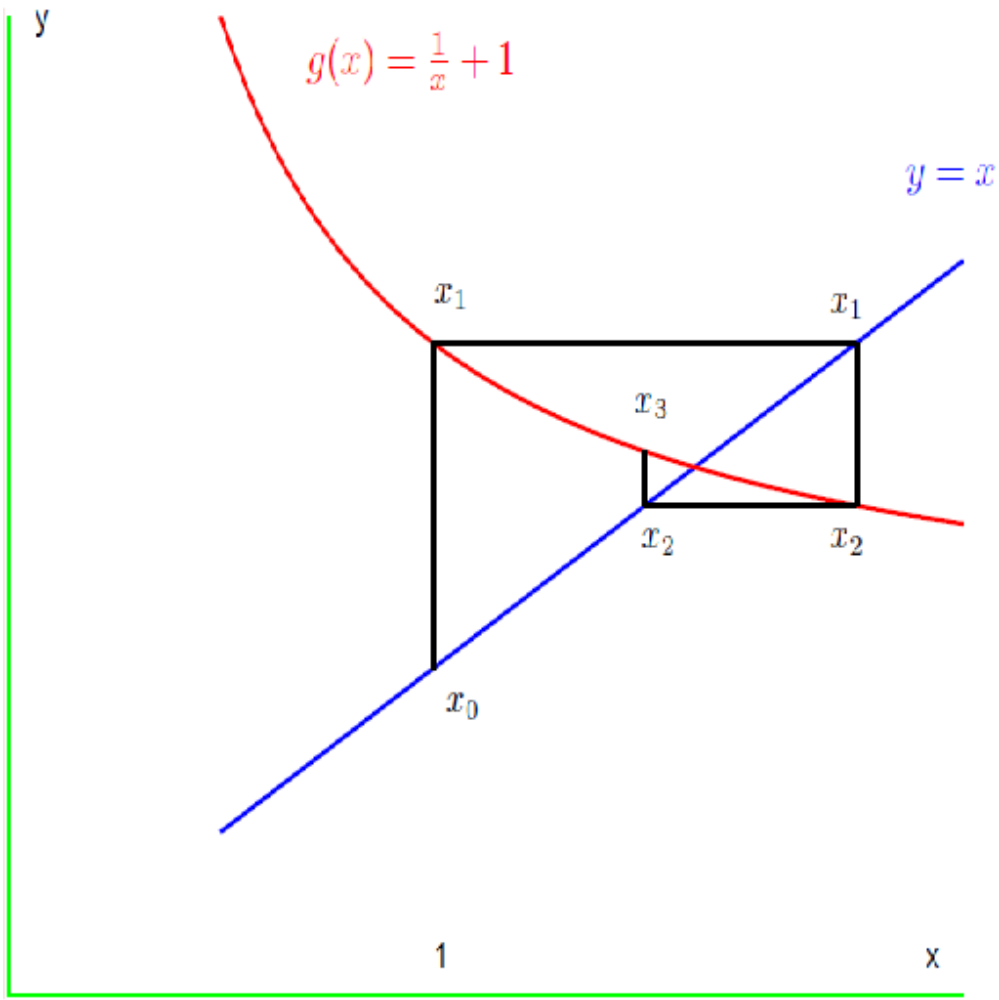
# Example 1

## Solution 2: $x_{n+1} = g(x_n) = 1 + \frac{1}{x_n}$ with $x_0 = 1$



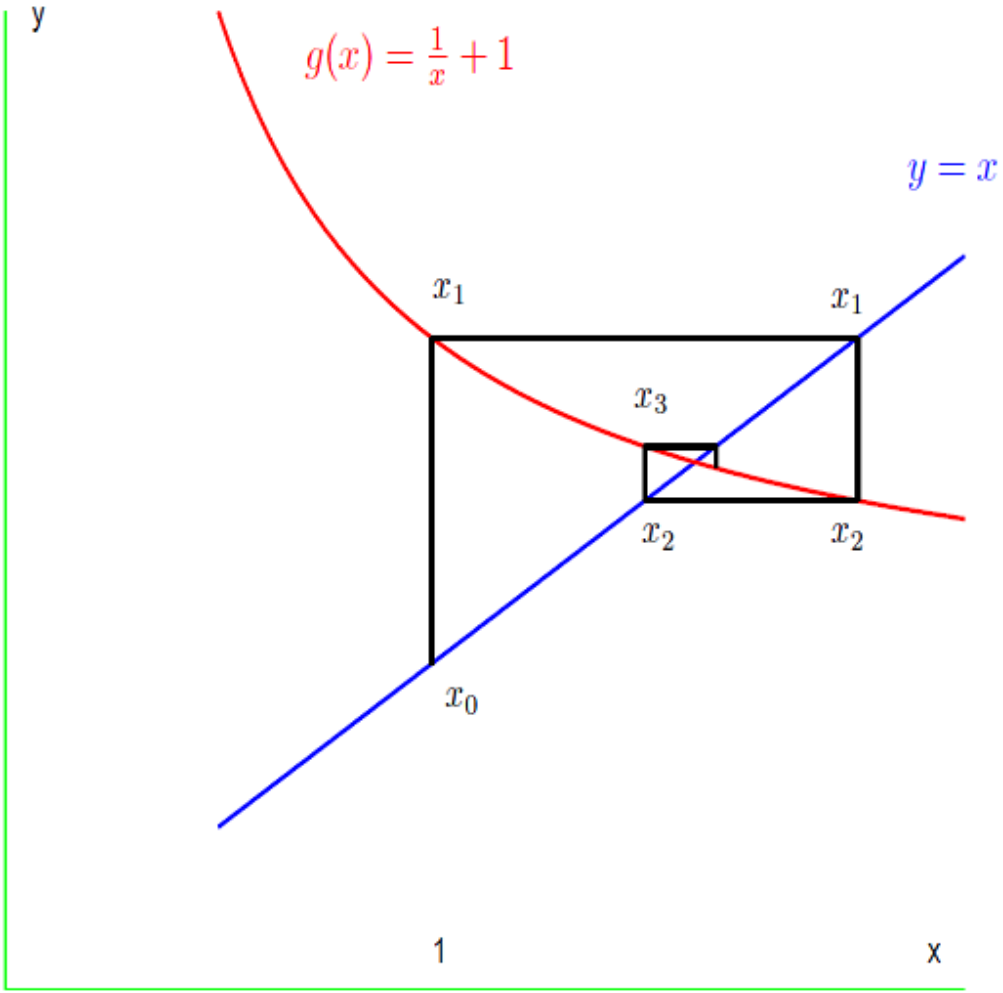
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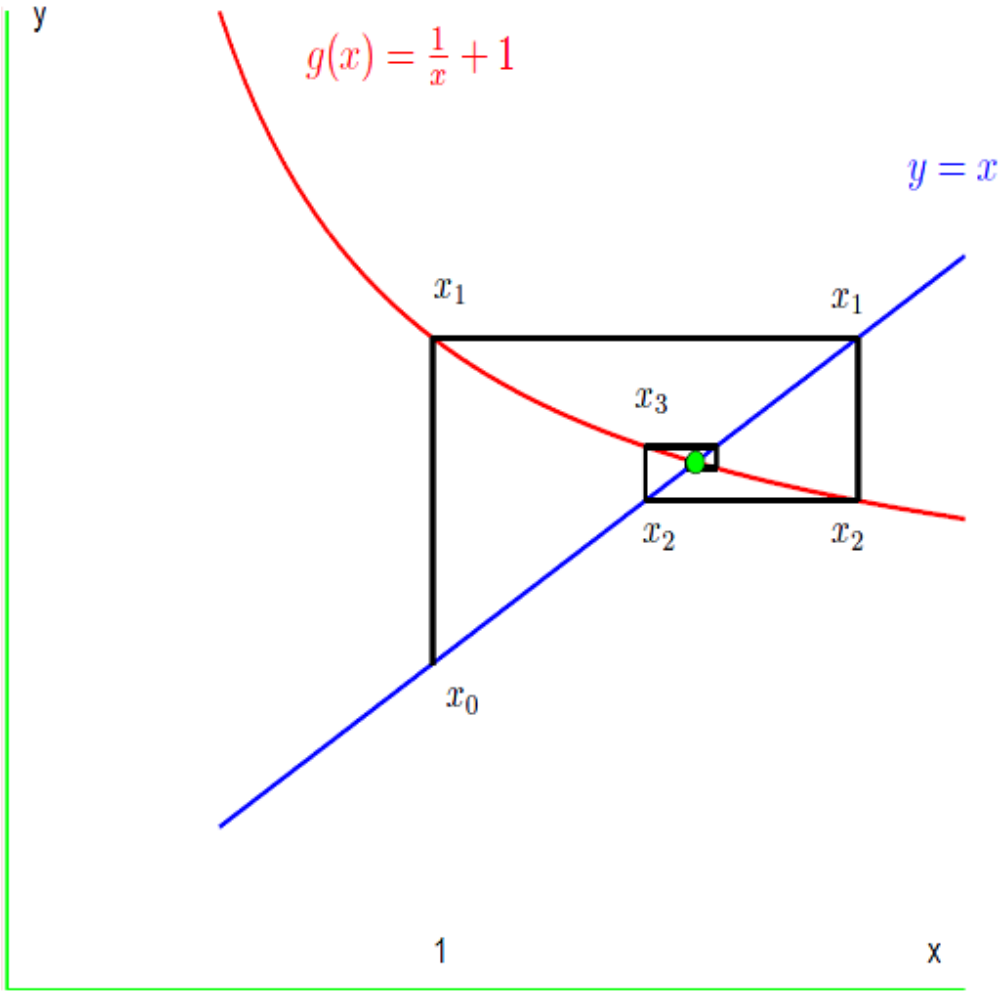
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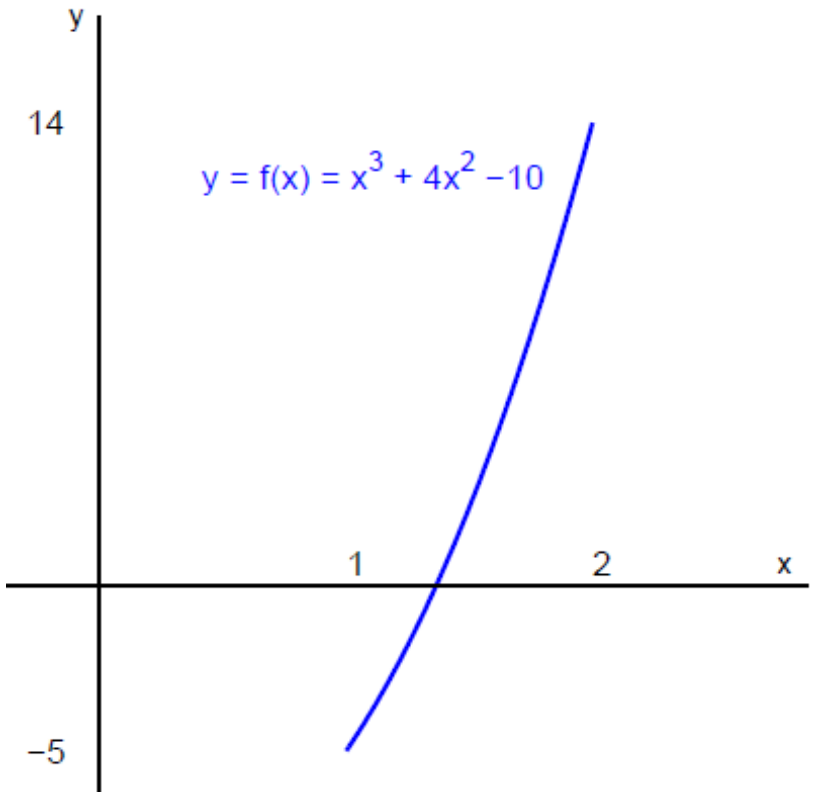


# Examples

## Example 2

Find the root for the equation:

$$x^3 + 4x^2 - 10 = 0$$



# Example 2

## Solutions: $x = g(x)$ with $x_0 = 1.5$

$x = g_1(x) = x - x^3 - 4x^2 + 10$ Does not Converge

$x = g_2(x) = \sqrt{\frac{10}{x} - 4x}$ Does not Converge

$x = g_3(x) = \frac{1}{2}\sqrt{10 - x^3}$ Converges after 31 Iterations

$x = g_4(x) = \sqrt{\frac{10}{4 + x}}$ Converges after 12 Iterations

$x = g_5(x) = x - \frac{x^3 + 4x^2 - 10}{3x^2 + 8x}$ Converges after 5 Iterations

# Fixed-Point Theorem

## Theorem (Fixed-Point Theorem)

*Let $g \in C[a,b]$ with $g(x) \in [a,b]$ for all $x \in [a,b]$. Let $g'(x)$ exist on $(a,b)$ with*

$$|g'(x)| \leq k < 1, \forall x \in [a,b].$$

*Then for any point $p_0$ in $[a,b]$, the sequence defined by*

$$p_n = g(p_{n-1}), n \geq 1$$

*will converge to the unique fixed point $p$ in $[a,b]$.*

## Proof

- By the Uniqueness Theorem, a unique fixed point exists in $[a,b]$.

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*will converge to the unique fixed point $p$ in $[a,b]$.*

## Proof

- By the Uniqueness Theorem, a unique fixed point exists in $[a,b]$.
- Since $g$ maps $[a,b]$ into itself, the sequence $\{p_n\}_{n=0}^{\infty}$ is defined for all $n \geq 0$ and $p_n \in [a,b]$ for all n.

# Fixed-Point Theorem

## Proof

- Using the MVT and the assumption that $|g'(x)| \leq k < 1, \forall x \in [a,b]$, we have

$$\begin{aligned} |p_n - p| = |g(p_{n-1}) - g(p)| &= |g'(\xi)||p_{n-1} - p| \\ &\leq k|p_{n-1} - p| \\ &\leq k^2|p_{n-2} - p| \\ &\leq k^n|p_0 - p| \end{aligned}$$

where $\xi \in (a,b)$.

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  where $\xi \in (a, b)$.

- Since $k < 1$,

$$\lim_{n\to\infty} |p_n - p| \leq \lim_{n\to\infty} k^n|p_0 - p| = 0,$$

  and $\{p_n\}_{n=0}^{\infty}$ converges to $p$.

# Fixed-Point Theorem

## Corollary (Corrollary to the Fixed-Point Theorem)

*If $g$ satisfies the hypothesis of the Theorem, then*

$$|p_n - p| \leq \frac{k^n}{1-k}|p_1 - p_0|.$$

## Proof

- $|p_{n+1} - p_n| = |g(p_n) - g(p_{n-1})| \leq k^n|p_1 - p_0|$

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- $|p_{n+1} - p_n| = |g(p_n) - g(p_{n-1})| \leq k^n|p_1 - p_0|$
- Thus, for $m > n \geq 1$

$$\begin{aligned}
|p_m - p_n| &= |p_m - p_{m-1} + p_{m-1} - p_{m-2} + \cdots + p_{n+1} - p_n| \\
&\leq |p_m - p_{m-1}| + |p_{m-1} - p_{m-2}| + \cdots + |p_{n+1} - p_n| \\
&\leq k^{m-1}|p_1 - p_0| + k^{m-2}|p_1 - p_0| + \cdots + k^n|p_1 - p_0| \\
&\leq k^n(1 + k + k^2 + \cdots + k^{m-n-1})|p_1 - p_0|.
\end{aligned}$$

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$$\begin{aligned} |p_m - p_n| &= |p_m - p_{m-1} + p_{m-1} - p_{m-2} + \cdots + p_{n+1} - p_n| \\ &\leq |p_m - p_{m-1}| + |p_{m-1} - p_{m-2}| + \cdots + |p_{n+1} - p_n| \\ &\leq k^{m-1}|p_1 - p_0| + k^{m-2}|p_1 - p_0| + \cdots + k^n |p_1 - p_0| \\ &\leq k^n(1 + k + k^2 + \cdots + k^{m-n-1})|p_1 - p_0|. \end{aligned}$$

- Since $\lim_{m\to\infty} p_m = p$, we obtain

$$|p - p_n| = \lim_{m\to\infty} |p_m - p_n| \leq k^n(1+k+k^2+\cdots+k^{m-n-1})|p_1 - p_0| = \frac{k^n}{1-k}|p_1 - p_0|.$$

# Example 2

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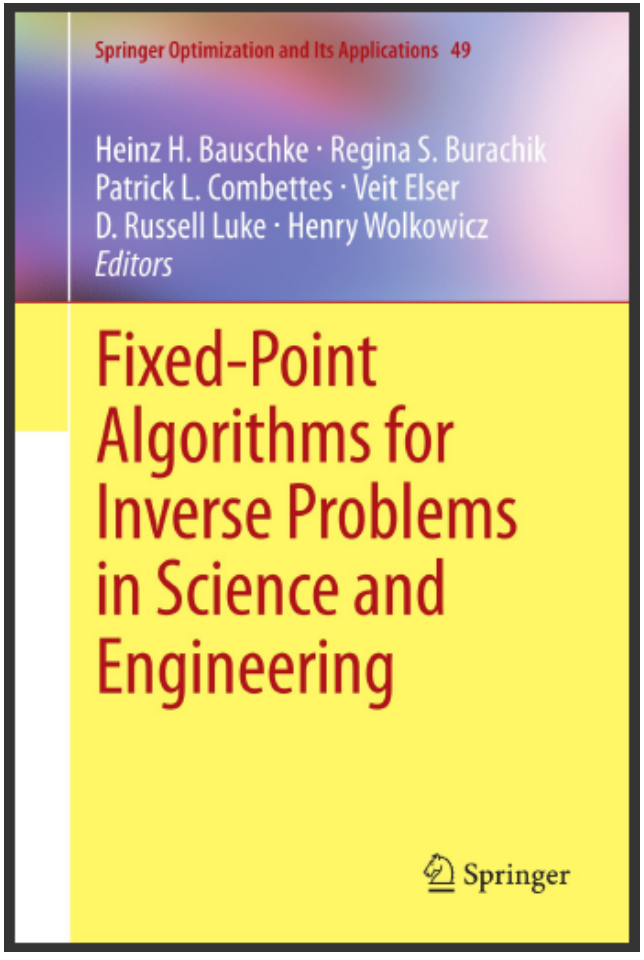
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# Question

How to construct a fixed-point problem?

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# Outline


1. The Root-Finding Problem
2. Newton's Method
3. Error Analysis for Iterative Methods

# Newton's Method

## Newton's Method

Newton's (or the Newton-Raphson) method is one of the most powerful and well-known numerical methods for solving a root-finding problem.

## Remarks

- Newton's method obtains faster convergence than offered by other types of functional iteration.
- Using Taylor polynomials. We will see there that this particular derivation produces not only the method, but also a bound for the error of the approximation.

# Newton's Method

## Derivation

- Suppose that $f \in C^2[a,b]$. Let $p_0 \in [a,b]$ be an approximation to $p$ such that $f'(p_0) \neq 0$ and $|p - p_0|$ is small.
- Consider the first Taylor polynomial for $f(x)$ expanded about $p_0$ and evaluated at $x = p$.

$$f(p) = f(p_0) + (p - p_0)f'(p_0) + \frac{(p - p_0)^2}{2}f''(\xi(p)),$$

where $\xi(p)$ lies between $p$ and $p_0$.

- Since $f(p) = 0$, this equation gives

$$0 = f(p_0) + (p - p_0)f'(p_0) + \frac{(p - p_0)^2}{2}f''(\xi(p)),$$

# Newton's Method

$$0 = f(p_0) + (p - p_0)f'(p_0) + \frac{(p - p_0)^2}{2}f''(\xi(p)),$$

## Derivation (cont'd)

- Newton's method is derived by assuming that since $|p - p_0|$ is small, the term involving $(p - p_0)^2$ is much smaller, so

$$0 \approx f(p_0) + (p - p_0)f'(p_0).$$

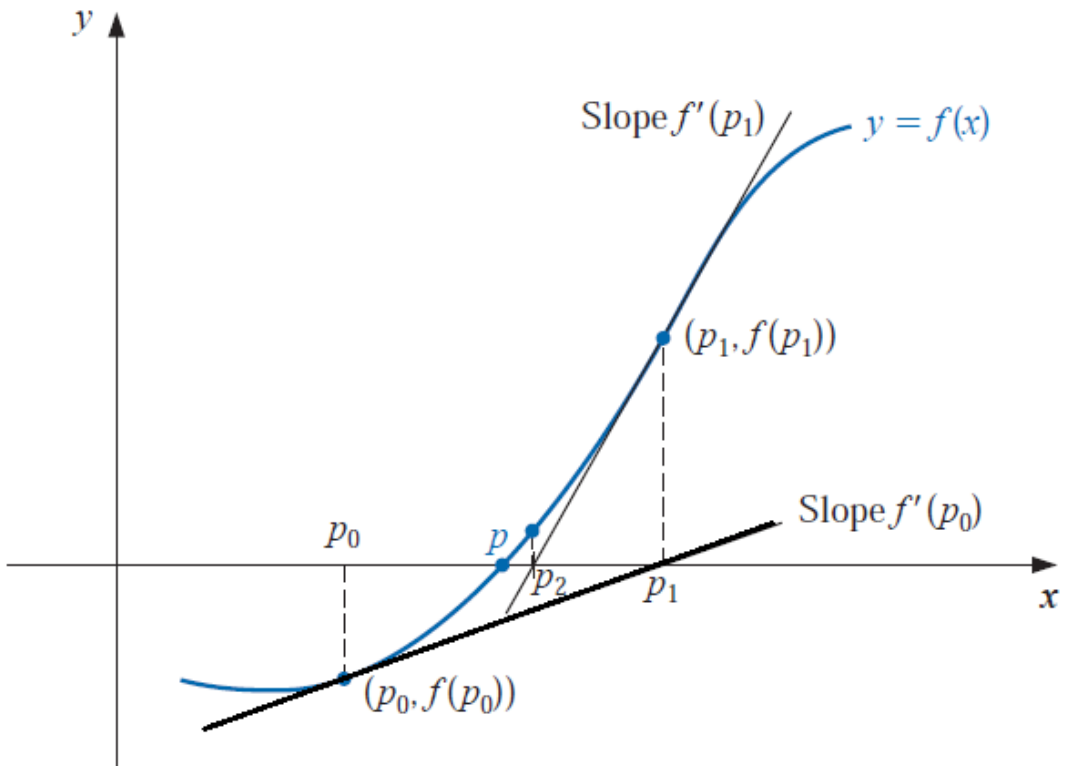
- Solving for $p$ gives

$$p \approx p_0 - \frac{f(p_0)}{f'(p_0)} \equiv p_1.$$

## Newton's Method

Starts with an initial approximation $p_0$ and generates the sequence $\{p_n\}_{n=0}^{\infty}$, by

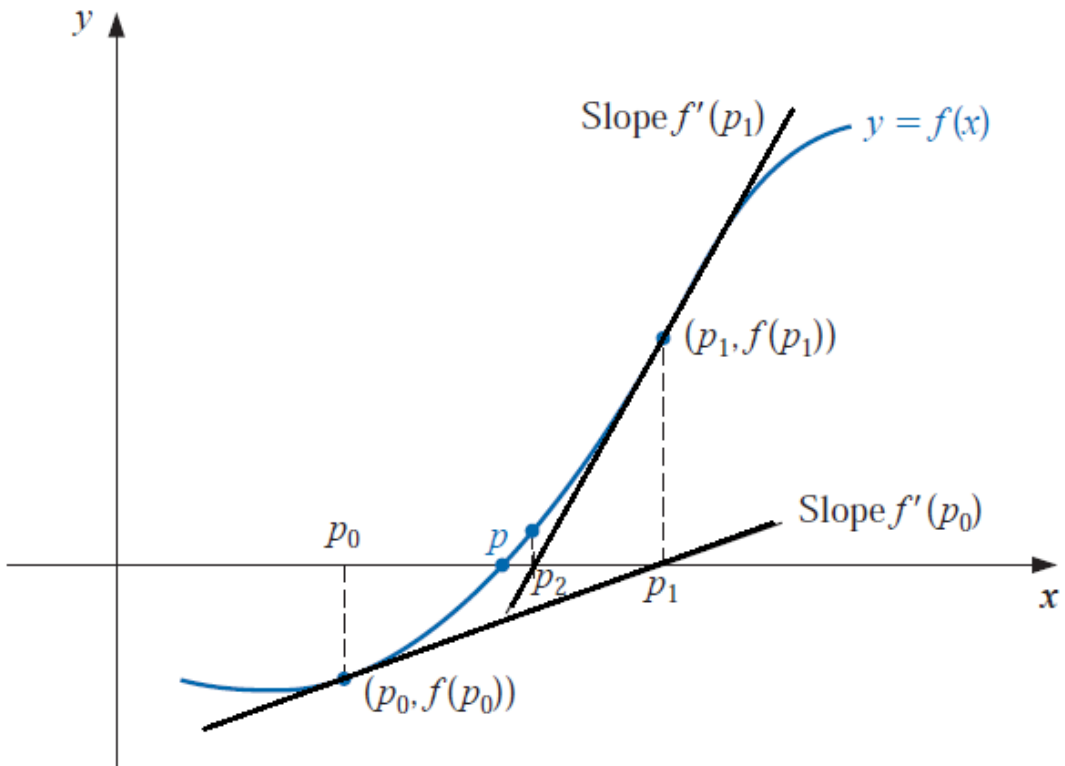
$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})} \qquad \text{for } n \geq 1$$



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Starts with an initial approximation $p_0$ and generates the sequence $\{p_n\}_{n=0}^{\infty}$, by

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})} \qquad \text{for } n \geq 1$$



## Newton's Algorithm

**INPUT** initial approximation $p_0$; tolerance $TOL$; maximum number of iterations $N_0$.

**OUTPUT** approximate solution $p$ or message of failure.

*Step 1* Set $i = 1$.

*Step 2* While $i \leq N_0$ do Steps 3–6.

*Step 3* Set $p = p_0 - f(p_0)/f'(p_0)$. (*Compute* $p_i$.)

*Step 4* If $|p - p_0| < TOL$ then
OUTPUT ($p$); (*The procedure was successful.*)
STOP.

*Step 5* Set $i = i + 1$.

*Step 6* Set $p_0 = p$. (*Update* $p_0$.)

*Step 7* OUTPUT ('The method failed after $N_0$ iterations, $N_0 =$', $N_0$);
(*The procedure was unsuccessful.*)
STOP.

# Newton's Algorithm

## Stopping Criteria for the Algorithm

- Various stopping criteria can be applied.
- We can select a tolerance $\epsilon > 0$ and generate $p_1, \cdots, p_N$ until one of the following conditions is met:

$$|p_N - p_{N-1}| < \epsilon \tag{1}$$

$$\frac{|p_N - p_{N-1}|}{p_N} < \epsilon, \quad p_N \neq 0, \quad \text{or} \tag{2}$$

$$|f(p_N)| < \epsilon \tag{3}$$

- Note that none of these inequalities give precise information about the actual error $|p_N - p|$.

# Newton's Method vs. Fixed-point Iteration

## Fixed-Point Iteration (a.k.a Functional Iteration)

- Newton's Method

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})} \qquad \text{for } n \geq 1$$

# Newton's Method vs. Fixed-point Iteration

## Fixed-Point Iteration (a.k.a Functional Iteration)

- Newton's Method

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})} \qquad \text{for } n \geq 1$$

- It can be written in the form

$$p_n = g(p_{n-1})$$

with

$$g(p_{n-1}) = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})} \qquad \text{for } n \geq 1$$

# Newton's Method vs. Fixed-point Iteration

## Example: Newton's Method vs. Fixed-point Iteration

Consider the function

$$f(x) = cos(x) - x = 0$$

Approximate a root of $f$ using

1. a fixed-point method
2. Newton's method

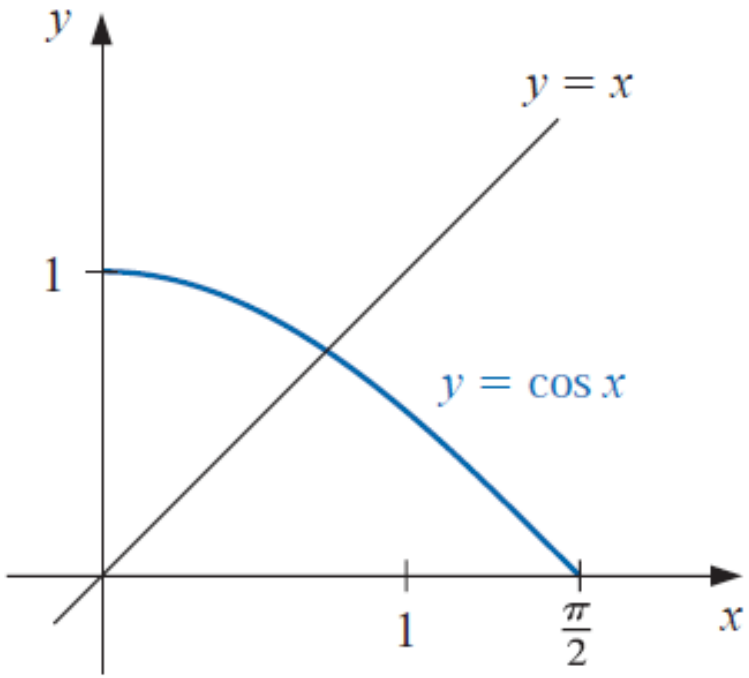
# Newton's Method vs. Fixed-point Iteration

## (1) Fixed-point Iteration for $f(x) = cos(x) - x$

- A solution to this root-finding problem is also a solution to the fixed-point problem

$$x = cos(x)$$

and the graph implies that a single fixed-point $p$ lies in $[0, \pi/2]$.



# Newton's Method vs. Fixed-point Iteration

## (1) Fixed-point Iteration: $x = cos(x)$, $x_0 = \pi/4$

- The following table shows the results of fixed-point iteration with $p_0 = \pi/4$.

| $n$ | $p_{n-1}$ | $p_n$ | $\lvert p_n - p_{n-1}\rvert$ | $e_n/e_{n-1}$ |
|---|---|---|---|---|
| 1 | 0.7853982 | 0.7071068 | 0.0782914 | — |
| 2 | 0.707107 | 0.760245 | 0.053138 | 0.678719 |
| 3 | 0.760245 | 0.724667 | 0.035577 | 0.669525 |
| 4 | 0.724667 | 0.748720 | 0.024052 | 0.676064 |
| 5 | 0.748720 | 0.732561 | 0.016159 | 0.671826 |
| 6 | 0.732561 | 0.743464 | 0.010903 | 0.674753 |
| 7 | 0.743464 | 0.736128 | 0.007336 | 0.672816 |

- The best conclusion from these results is that $p \approx 0.74$.

# Newton's Method vs. Fixed-point Iteration

## (2) Newton's Method for $f(x) = cos(x) - x$

- To apply Newton's method to this problem we need

$$f'(x) = -sin(x) - 1$$

- Starting with $p_0 = \pi/4$, we generate the sequence defined for $n \geq 1$, by

$$\begin{aligned} p_n &= p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})} \\ &= p_{n-1} - \frac{cos(p_{n-1}) - p_{n-1}}{-sin(p_{n-1}) - 1}. \end{aligned}$$

# Newton's Method vs. Fixed-point Iteration

## (2) Newton's Method for $f(x) = cos(x) - x$, $x_0 = \pi/4$

- The following table shows the results of Newton's method with $p_0 = \pi/4$.

| $n$ | $p_{n-1}$ | $f(p_{n-1})$ | $f'(p_{n-1})$ | $p_n$ | $\lvert p_n - p_{n-1} \rvert$ |
|---|---|---|---|---|---|
| 1 | 0.78539816 | -0.078291 | -1.707107 | 0.73953613 | 0.04586203 |
| 2 | 0.73953613 | -0.000755 | -1.673945 | 0.73908518 | 0.00045096 |
| 3 | 0.73908518 | -0.000000 | -1.673612 | 0.73908513 | 0.00000004 |
| 4 | 0.73908513 | -0.000000 | -1.673612 | 0.73908513 | 0.00000000 |

- An excellent approximation is obtained with $n = 3$.
- Because of the agreement of $p_3$ and $p_4$ we could reasonably expect this result to be accurate to the places listed.

# Convergence using Newton's Method

## Convergence Theorem for Newton's Method

Let $f \in C^2[a,b]$. If $p \in (a,b)$ is such that $f(p) = 0$ and $f'(p) \neq 0$. Then there exists a $\delta > 0$ such that Newton's method generates a sequence $\{p_n\}_{n=1}^{\infty}$, defined by

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}$$

converging to $p$ for any initial approximation

$$p_0 \in [p - \delta, p + \delta].$$

# Convergence using Newton's Method

## Proof: (1/4)

- The proof is based on analyzing Newton's method as the functional iteration scheme $p_n = g(p_{n-1})$, for $n \geq 1$ with

$$g(x) = x - \frac{f(x)}{f'(x)}.$$

- Let $k \in (0, 1)$. We find an interval $[p - \delta, p + \delta]$ that $g$ maps into itself and for which $g'(x) \leq k$, for all $x \in (p - \delta, p + \delta)$.
- Since $f'$ is continuous and $f'(p) \neq 0$, there exists a $\delta_1 > 0$, such that $f'(x) \neq 0$ for $x \in [p - \delta_1, p + \delta_1] \subseteq [a, b]$.

# Convergence using Newton's Method

## Proof: (2/4)

- Thus $g$ is defined and continuous on $[p-\delta_1, p+\delta_1]$. Also

$$g'(x) = 1 - \frac{f'(x)f'(x) - f(x)f''(x)}{[f'(x)]^2} = \frac{f(x)f''(x)}{[f'(x)]^2},$$

  for $x \in [p-\delta_1, p+\delta_1]$, and since $f \in C^2[a,b]$, we have $g \in C^1[p-\delta_1, p+\delta_1]$.

- By assumption, $f(p) = 0$, so

$$g'(p) = \frac{f(p)f''(p)}{[f'(p)]^2} = 0.$$

# Convergence using Newton's Method

$$g'(p) = \frac{f(p)f''(p)}{[f'(p)]^2} = 0.$$

## Proof: (3/4)

- Since $g'$ is continuous and $0 < k < 1$, there exists a $\delta$, with $0 < \delta < \delta_1$, and

$$|g'(x)| \leq k, \quad \text{for all} \quad x \in [p - \delta, p + \delta].$$

- It remains to show that $g$ maps $[p - \delta, p + \delta]$ into $[p - \delta, p + \delta]$.
- If $x \in [p - \delta, p + \delta]$, the MVT implies that for some number $\xi$ between $x$ and $p$, $|g(x) - g(p)| = |g'(\xi)||x - p|$. So

$$|g(x) - p| = |g(x) - g(p)| = |g'(\xi)||x - p| \leq k|x - p| < |x - p|.$$

# Convergence using Newton's Method

## Proof: (4/4)

- Since $x \in [p-\delta, p+\delta]$, it follows that $|x-p| < \delta$ and that $|g(x)-p| < \delta$. Hence, $g$ maps $[p-\delta, p+\delta]$ into $[p-\delta, p+\delta]$.
- All the hypotheses of the Fixed-Point Theorem are now satisified, so the sequence $\{p_n\}_{n=1}^{\infty}$, defined by

$$p_n = p_{n-1} - \frac{f(p_{n-1})}{f'(p_{n-1})}$$

converges to $p$ for any $p_0 \in [p-\delta, p+\delta]$.

# Remarks of Newton's Method

## Choice of Initial Approximation

- The convergence theorem states that, under reasonable assumptions, Newton's method converges if a sufficiently accurate initial approximation is chosen.
- It also implies that the constant $k$ that bounds the derivative of $g$, and consequently, indicates the speed of convergence of the method, decreases to 0 as the procedure continues.
- This result is important for the theory of Newton's method, but it is seldom applied in practice because it does not tell us how to determine $\delta$.

# Remarks of Newton's Method

## In a practical application ...

- an initial approximation is selected
- and successive approximations are generated by Newton's method.
- These will generally either converge quickly to the root, or it will be clear that convergence is unlikely.

# Remarks of Newton's Method

## Weakness of Newton's Method

- It needs to know the value of the derivate of $f$ at each approximation.
- Frequently, $f'(x)$ is far more difficult and needs more arithmetic operations to calculate than $f(x)$.

# Outline

1 The Root-Finding Problem

2 Newton's Method

3 Error Analysis for Iterative Methods

# Error Analysis for Iterative Methods

## Order of Convergence

Suppose $\{p_n\}_{n=0}^{\infty}$ is a sequence that converges to $p$, with $p_n \neq p$ for all $n$. If positive constants $\lambda$ and $\alpha$ exist with

$$\lim_{n\to\infty} \frac{|p_{n+1} - p|}{|p_n - p|^{\alpha}} = \lambda$$

then $\{p_n\}_{n=0}^{\infty}$ converges to $p$ of order $\alpha$, with asymptotic error constant $\lambda$.

- If $\alpha = 1$, the sequence is linearly convergent.
- If $\alpha = 2$, the sequence is quadratically convergent.

# Error Analysis for Iterative Methods

## linearly convergent vs. quadratically convergent

Suppose that $\{p_n\}_{n=0}^{\infty}$ is linearly convergent to 0 with

$$\lim_{n\to\infty} \frac{|p_{n+1}|}{|p_n|} = 0.5$$

and that $\{\tilde{p}_n\}_{n=0}^{\infty}$ is quadratically convergent to 0 with the same asymptotic error constant,

$$\lim_{n\to\infty} \frac{|\tilde{p}_{n+1}|}{|\tilde{p}_n|^2} = 0.5.$$

For simplicity we assume that for each $n$ we have

$$\frac{|p_{n+1}|}{|p_n|} \approx 0.5 \quad \text{and} \quad \frac{|\tilde{p}_{n+1}|}{|\tilde{p}_n|^2} \approx 0.5.$$

For the linearly convergent scheme, this means that

$$|p_n - 0| = |p_n| \approx 0.5|p_{n-1}| \approx (0.5)^2|p_{n-2}| \approx \cdots \approx (0.5)^n|p_0|,$$

whereas the quadratically convergent procedure has

$$\begin{aligned}|\tilde{p}_n - 0| = |\tilde{p}_n| &\approx 0.5|\tilde{p}_{n-1}|^2 \approx (0.5)[0.5|\tilde{p}_{n-2}|^2]^2 = (0.5)^3|\tilde{p}_{n-2}|^4 \\ &\approx (0.5)^3[(0.5)|\tilde{p}_{n-3}|^2]^4 = (0.5)^7|\tilde{p}_{n-3}|^8 \\ &\approx \cdots \approx (0.5)^{2^n-1}|\tilde{p}_0|^{2^n}.\end{aligned}$$

# Error Analysis for Iterative Methods

linearly convergent vs. quadratically convergent

| $n$ | Linear Convergence Sequence $\{p_n\}_{n=0}^{\infty}$ $(0.5)^n$ | Quadratic Convergence Sequence $\{\tilde{p}_n\}_{n=0}^{\infty}$ $(0.5)^{2^n-1}$ |
|---|---|---|
| 1 | $5.0000 \times 10^{-1}$ | $5.0000 \times 10^{-1}$ |
| 2 | $2.5000 \times 10^{-1}$ | $1.2500 \times 10^{-1}$ |
| 3 | $1.2500 \times 10^{-1}$ | $7.8125 \times 10^{-3}$ |
| 4 | $6.2500 \times 10^{-2}$ | $3.0518 \times 10^{-5}$ |
| 5 | $3.1250 \times 10^{-2}$ | $4.6566 \times 10^{-10}$ |
| 6 | $1.5625 \times 10^{-2}$ | $1.0842 \times 10^{-19}$ |
| 7 | $7.8125 \times 10^{-3}$ | $5.8775 \times 10^{-39}$ |

# Error Analysis for Iterative Methods

## Theorem (Fixed Point Method)

*Let $g \in C[a,b]$ be such that $g(x) \in [a,b]$ for all $x \in [a,b]$. Suppose, in addition, that $g'(x)$ is continuous on $(a,b)$ and a positive constant $k < 1$ exists with*

$$|g'(x)| \leq k, \forall x \in [a,b].$$

*If $g'(p) \neq 0$, then for any number $p_0 \neq p$ in $[a,b]$, the sequence*

$$p_n = g(p_{n-1}), n \geq 1$$

*converges only linearly to the unique fixed point $p$ in $[a,b]$.*

# Error Analysis for Iterative Methods

## Proof

- Since $g'$ exists on $(a, b)$, applying the MVT, we have

$$p_{n+1} - p = g(p_n) - g(p) = g'(\xi_n)(p_n - p)$$

- Since $\{p_n\}_{n=0}^{\infty}$ converges to $p$, $\{\xi_n\}_{n=0}^{\infty}$ also converges to $p$.
- Thus,

$$\lim_{n\to\infty} \frac{p_{n+1} - p}{p_n - p} = \lim_{n\to\infty} g'(\xi_n) = g'(p)$$

Hence, if $g'(p) \neq 0$, fixed-point iteration exhibits linear convergence with asymptotic error constant $|g'(p)|$.

# Error Analysis for Iterative Methods

## Theorem

*Let $p$ be a solution of the equation $x = g(x)$. Suppose that $g'(p) = 0$ and $g''$ is continuous with $g''(x) < M$ on an open interval $I$ containing $p$. Then there exists a $\delta > 0$ such that, for $p_0 \in [p - \delta, p + \delta]$, the sequence defined by $p_n = g(p_{n-1})$, when $n \geq 1$, converges at least quadratically to $p$. Moreover, for sufficiently large values of $n$,*

$$|p_{n+1} - p| < \frac{M}{2}|p_n - p|^2.$$

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## Proof (1/2)

- Choose $k$ in $(0, 1)$ and $\delta > 0$ such that on the interval $[p - \delta, p + \delta]$ contained in $I$, we have $|g'(x)| \leq k$ and$g''$ continuous.
- Expanding $g(x)$ in a linear Taylor polynomial for $x \in [p - \delta, p + \delta]$ gives

$$g(x) = g(p) + g'(p)(x - p) + \frac{g''(\xi)}{2}(x - p)^2$$

- The hypotheses $g(p) = p$ and $g'(p) = 0$ imply that

$$g(x) = p + \frac{g''(\xi)}{2}(x - p)^2$$

## Proof (2/2)

- When $x = p_n$, we have

$$p_{n+1} = g(p_n) = p + \frac{g''(\xi)}{2}(p_n - p)^2$$

- Thus,

$$\lim_{n\to\infty} \frac{|p_{n+1} - p|}{|p_n - p|^2} = \frac{|g''(p)|}{2} < \frac{M}{2}$$

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## Remarks

- For a fixed point method to converge quadratically we need to have both $g(p) = p$, and $g'(p) = 0$.
- If $f(p) = 0$ and $f'(p) \neq 0$, then for starting values sufficiently close to $p$, Newton's method will converge at least quadratically.

# Convergence using Newton's Method

## Proof: (2/4)

- Thus $g$ is defined and continuous on $[p-\delta_1, p+\delta_1]$. Also

$$g'(x) = 1 - \frac{f'(x)f'(x) - f(x)f''(x)}{[f'(x)]^2} = \frac{f(x)f''(x)}{[f'(x)]^2},$$

  for $x \in [p-\delta_1, p+\delta_1]$, and since $f \in C^2[a,b]$, we have $g \in C^1[p-\delta_1, p+\delta_1]$.

- By assumption, $f(p) = 0$, so

$$g'(p) = \frac{f(p)f''(p)}{[f'(p)]^2} = 0.$$

# Assignment

- Numerical Analysis, Chapter 1 & 2;
- Homework 1.